Jadavpur University
Department of Computer Science and Engineering
M. Tech in Computer Technology
1st Year, 1st Semester
Machine Learning and Data Science Lab
Assignment

Group-A

A.1. Practicing with Numpy arrays
  a) Printing the type of arr object
  b) Printing array dimensions
  c) Printing the shape of the array
  d) Printing size (total number of elements) of the array
  e) Printing the type of elements in an array
  f) Slicing of a given array

A.2. Write a program to create a data frame using Pandas

A.3. Write a Python program for Basic plots using Matplotlib

A.4. Write a Python program to compute Frequency distributions

A.5. Write a Python program to calculate the average and variance

A.6. Write a Python program to draw Normal Curves

A.7. Write a Python program to draw Scatter plots

A.8. Write a Python program to compute a Correlation coefficient

Group-B

B.1. Write Python code to implement a regression model for weather forecasting. Compare simple regression, Ridge regression, and Lasso regression.

B.2. Write Python code to implement a Logistic Regression Model for Spam detection.

B.3. Write Python code to implement the ID3 algorithm and test it on the PlayTennis dataset, verifying the inductive bias of the decision tree learning algorithm.

B.4. Implement the K-Means algorithm and visualize the data in 2-D space before clustering. Compare the performance of the K-means clustering algorithm with the DBSCAN algorithm.

B.5. Implement the spectral clustering algorithm and test its performance on the spiral dataset. Compare these clustering results with those obtained by the simple K-means clustering algorithm.

B.6. Implement a hierarchical clustering algorithm and display the clustering results using a Dendrogram or a Venn Diagram.

B.7. Write Python code to implement the Backpropagation algorithm for an ANN with more than two hidden layers. Develop two such deep models with the following configurations.

Model 1: uses sigmoid activations at the hidden nodes and softmax activation function at the output nodes.

Model 2: uses ReLu activation function at the hidden nodes and softmax activation function at the output nodes.

The hyperparameters (e.g., learning rate, momentum, number of hidden layers, number of hidden nodes per layer) for both models should be properly tuned using a validation set. Compare the performance of these two models on the MNIST dataset when both are trained for 1000 epochs.

B.8. Implement KNN, Naïve Bayes, and SVM classifiers using Python-based ML tools for comparison of their performance on the review sentiment classification dataset.